FOOD SUPPLY CHAIN STOCK RECONCILIATION

Algorithm, Flowchart, Pseudocode and Sample Output

1. Algorithm

Start the program.

Display the title: Food Supply Chain Stock Reconciliation.

Input the number of locations.

Repeat the following steps for each location:

a. Input the location name and food item.

b. Input quantity received, quantity distributed, quantity damaged, actual remaining stock and target stock (in kg).

c. Calculate expected remaining stock = received - distributed - damaged.

d. Calculate discrepancy = actual remaining stock - expected remaining stock.

e. Calculate replenishment required = max(0, target stock - actual remaining stock).

f. Display the location, food item, expected stock, actual stock, discrepancy and replenishment required.

g. If discrepancy is negative, display a stock-shortage alert.

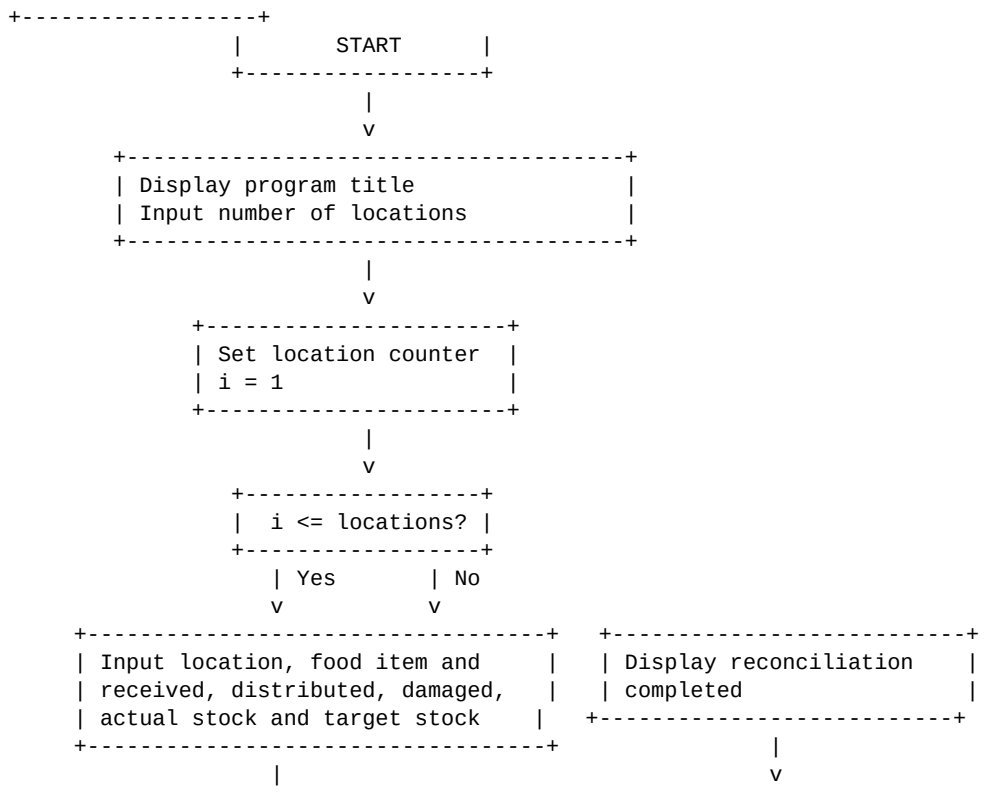
h. Otherwise, if discrepancy is positive, display an extra-stock alert.

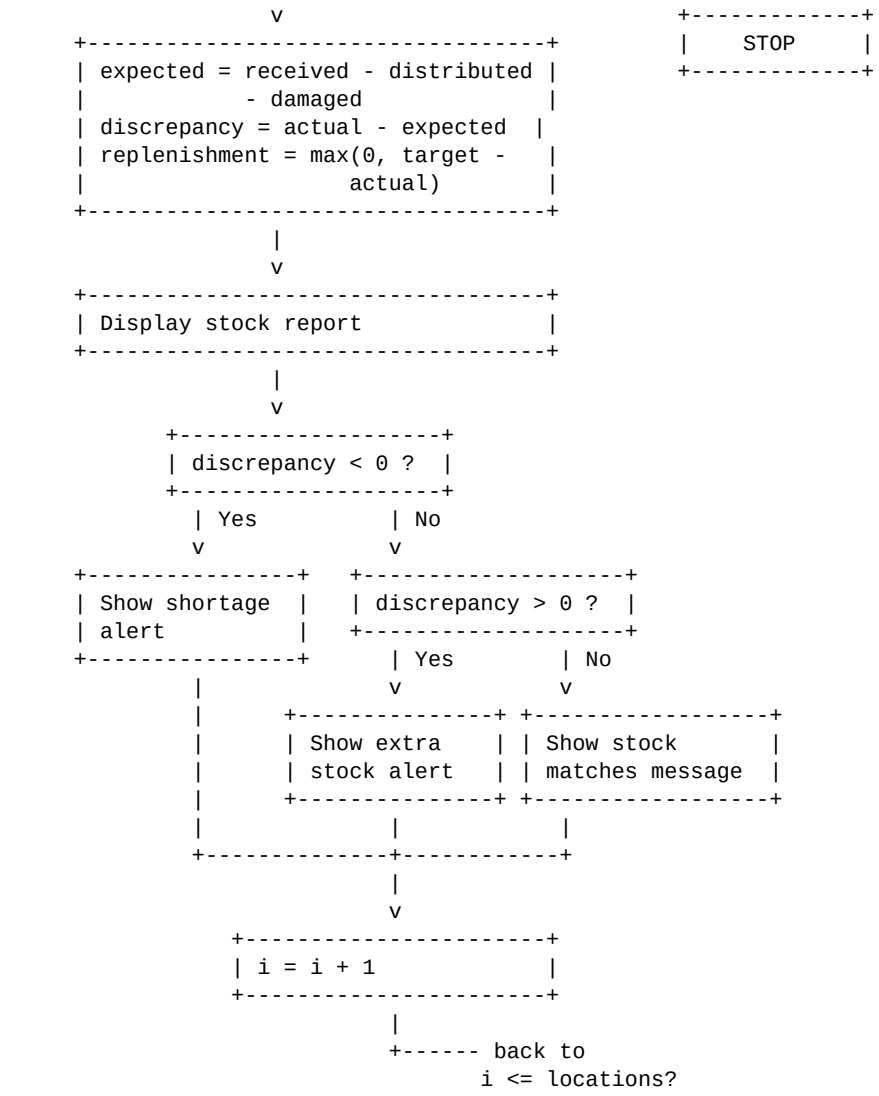
i. Otherwise, display that stock matches the expected quantity.

After processing all locations, display “Stock reconciliation completed!”.

Stop.

2. Flowchart





3. Pseudocode

```
BEGIN
    DISPLAY "FOOD SUPPLY CHAIN STOCK RECONCILIATION"

    INPUT locations

    FOR i = 1 TO locations
        DISPLAY "Location", i

        INPUT name
        INPUT food
        INPUT received
        INPUT distributed
        INPUT damaged
        INPUT actual
        INPUT target

        expected = received - distributed - damaged
        discrepancy = actual - expected
        replenishment = MAX(0, target - actual)

        DISPLAY "Stock Report"
        DISPLAY name, food
        DISPLAY expected, actual
        DISPLAY discrepancy
        DISPLAY replenishment

        IF discrepancy < 0 THEN
            DISPLAY "Alert: Stock shortage detected!"
```

```
        ELSE IF discrepancy > 0 THEN
            DISPLAY "Alert: Extra stock detected!"
        ELSE
            DISPLAY "Stock matches expected quantity."
        END IF
    END FOR

    DISPLAY "Stock reconciliation completed!"
END
```

4. Sample Output

```
FOOD SUPPLY CHAIN STOCK RECONCILIATION
--------------------------------------
Enter number of locations: 1

Location 1
Enter location name: Central Store
Enter food item: Rice
Enter quantity received (kg): 100
Enter quantity distributed (kg): 60
Enter quantity damaged (kg): 5
Enter actual remaining stock (kg): 35
Enter target stock (kg): 50

--- Stock Report ---
Location: Central Store
Food item: Rice
Expected remaining stock: 35 kg
Actual remaining stock: 35 kg
Discrepancy: 0 kg
Replenishment required: 15 kg
Stock matches expected quantity.

Stock reconciliation completed!
```

Note: The sample values are illustrative. Expected stock is calculated as received minus distributed minus damaged. The discrepancy is actual stock minus expected stock. Replenishment is the positive amount needed to reach the target, or zero if the actual stock already meets or exceeds the target.